IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MATHEMATICS 1A

Wednesday, 7 January 2026, 10:00 AM

Time allowed: 2:00 hours

There are FOUR questions on this paper.

This exam has TWO sections.
Section A (Question ONE) is worth 40 marks, and is multiple choice. Please enter your answers on the multiple-choice answer sheet. Please note that some of the questions are worth 2 marks, and some 3 marks.
Section B has THREE longer questions, worth 20 marks each. Enter your answers in the booklet provided.

NO CALCULATORS ALLOWED. Formula sheet will be provided.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) :

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ELEC40013 - MATHEMATICS1A - QUESTIONS

Section A: Multiple Choice

Enter your answers on the multiple-choice answer sheet. Please note that some of the questions are worth 2 marks, and some 3 marks. Total marks for this section: 40.

1. Given complex $z = x + iy$, which of the following are the real and imaginary parts of $w = \sin(iz)$? (See also Section B, Q1a.)

   (a) $\mathrm{Im}(w) = \sin y \cosh x$, and $\mathrm{Re}(w) = -\cos y \sinh x$,

   (b) $\mathrm{Re}(w) = -\cos y \cosh x$, and $\mathrm{Im}(w) = \sin y \sinh x$,

   (c) $\mathrm{Im}(w) = \sin x \cosh y$, and $\mathrm{Re}(w) = -\cos x \sinh y$,

   (d) $\mathrm{Re}(w) = -\sin y \cosh x$, and $\mathrm{Im}(w) = \cos y \sinh x$,

   (e) none of the above. [3]

2. Which of the following accurately describes the locus in the complex plane satisfying $|z-i| < |z+2|$?

   (a) the area above the line $y = -2x - \frac{3}{2}$, (b) the area below the line $y = 2x - \frac{1}{2}$,

   (c) the area below the line $x = -2y + \frac{1}{2}$, (d) the area above the line $x = 2y + \frac{3}{2}$,

   (e) none of the above. [3]

3. Which of the following statements about $\lim_{x\to\infty} \dfrac{\sqrt{2x+1} - \sqrt{2x-1}}{x^{-1/2}}$ is true?

   (a) the limit does not exist, (b) the limit is equal to 0,

   (c) the limit is equal to $\frac{1}{\sqrt{2}}$, (d) the limit cannot be determined,

   (e) none of the above. [3]

4. The limit given as $\lim_{x\to -1} \dfrac{1-\cos(x+1)}{(x+1)^2}$

   (a) exists, but cannot be determined, (b) is equal to 0,

   (c) does not exist, (d) is equal to $\frac{1}{2}$,

   (e) none of the above. [3]

5. Let two functions be $f(x) = \ln(x + \sqrt{x^2+1})$ and $g(x) = \dfrac{e^{-x}\ln x}{x}$. Then

(a) $f'(x) = \dfrac{1}{\sqrt{x^2+1}}$, and $\dfrac{dg}{dx} = \dfrac{e^{-x}[(\ln x - 1)x - 1)}{x^2}$,

(b) $\dfrac{dg}{dx} = \dfrac{1}{\sqrt{x^2+1}}$, and $f'(x) = \dfrac{e^{-x}[(1-(x+1)\ln x]}{x^2}$,

(c) $\dfrac{df}{dx} = \dfrac{1}{\sqrt{x^2-1}}$, and $g'(x) = \dfrac{e^{-x}[\ln x(x-1)+1]}{x^2}$,

(d) $f'(x) = \dfrac{x}{\sqrt{x^2+1}}$, and $g'(x) = \dfrac{e^{-x}(x - \ln x + 1)}{x^2}$,

(e) none of the above. [2]

6. Which of the following statements about the function $g(x) = \dfrac{e^{-x}\ln x}{x}$ is true? (See also Section B, Q1b.)

(a) there is a local maximum at x_1, between 0 and 1,

(b) as $x \to 0$, $g(x) \to \infty$, so the $y-$axis is an asymptote,

(c) as $x \to \infty$, $g(x) \to 0^-$, so the $x-$axis is an asymptote,

(d) there is a point of inflection at $x_2 > 1$

(e) none of the above. [2]

7. If $x = 1 - \cos v$, and $y = \sin v - v$, where v is a parameter, which of the following is $\dfrac{dy}{dx}$?

(a) $\cot\left(\dfrac{v}{2}\right)$, (b) $-\tan\left(\dfrac{v}{2}\right)$, (c) $\cot v + \csc v$, (d) $\dfrac{\cos v + 1}{\sin v}$,

(e) none of the above. [2]

8. If $I(x) = \displaystyle\int_0^x te^{-t}\,dt$ is a function of x, which of the following statements is **false**?

(a) $I(x)$ increases most rapidly when $x = 1$, (b) $I(x)$ has a point of inflection at $x = 2$,

(c) $I(x)$ has a stationary point at $x = 0$, (d) $I(x)$ is increasing for all $x > 0$,

(e) none of the above . [3]

9. The integral $I = \displaystyle\int \frac{1}{\cos x - 1}\,dx$ is equal to

(a) $\cot(x/2)+$C, (b) $\cot x - \csc x +$C, (c) $\dfrac{x}{\cos x + 1}+$C, (d) $\dfrac{1}{\tan(x/2) - 1}+$C,

(e) none of the above. [3]

10. Two series are given as

$$S_1 = \sum_{n=1}^{\infty} \frac{x^n}{n} \quad \text{and} \quad S_2 = \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}} .$$

If R means radius of convergence, which of the following statements is **false**?

(a) S_1 has $R = 1$, and S_2 converges conditionally,

(b) S_2 is an alternating series, and S_1 converges for $x = -1$,

(c) S_1 diverges for $x = 1$, and S_2 has $R = 1$,

(d) S_2 converges, but not by the integral test, and S_1 diverges for $|x| > 1$,

(e) none of the above. [3]

11. Which of the following are the first three terms of the power series of $f(x) = \sqrt{1-x}$?

(a) $1 - \frac{1}{2}x - \frac{1}{4}x^2$, (b) $1 - \frac{1}{2}x - \frac{1}{8}x^2$, (c) $1 + \frac{1}{2}x + \frac{1}{4}x^2$, (d) $1 - \frac{1}{2}x + \frac{1}{8}x^2$,

(e) none of the above. [2]

12. In the previous question, we obtained the first three terms of the Maclaurin series of $f(x) = \sqrt{1-x}$. Which of the following is the correct remainder term R_2?

(a) $R_2 = -\frac{x^3}{16(1-c)^{5/2}}$, with $0 < c < x$, or $x < c < 0$,

(b) $R_2 = -\frac{1}{4(1+c)^{3/2}}x^2$, with $c < x < 0$ or $x > c > 0$,

(c) $R_2 = -\frac{3x^3}{8(1+c)^{5/2}}$, with $0 < |c| < x$,

(d) $R_2 = \frac{1}{8(1-c)^{3/2}}x^2$, with $0 < x < c$ or $x > c > 0$,

(e) none of the above. [3]

13. To solve the first order ordinary differential equation (ODE)

$$x^2\frac{dy}{dx} - xy + y^2 = 0$$

we begin by rewriting in terms of $v = \frac{y}{x}$, to obtain the following separable first-order ODE:

(a) $\frac{dv}{dx} = -v^2 + v$, (b) $x\frac{dv}{dx} = -v^2$, (c) $-\frac{1}{v - v^2}\frac{dv}{dx} = \frac{1}{x}$, (d) $\frac{1}{x}\frac{dv}{dx} = -\frac{1}{v^2}$,

(e) none of the above . [3]

14. The solution of the homogenous first-order ODE

$$x^2\frac{dy}{dx} - xy + y^2 = 0$$

is obtained from the separable ODE found in the previous question, with $C \in \mathbb{R}$, as follows:

(a) $y = -\dfrac{x}{C+\ln x}$, (b) $y = \dfrac{\ln(Cx)}{x}$, (c) $y = \dfrac{x}{\ln(Cx)}$, (d) $y = -x(C+\ln x)$,

(e) none of the above. [3]

15. A function of two variables is given as $f(x,y) = e^{xy}\sin x$. Which of the following are correct first partial derivatives?

(a) $f_x = xe^{xy}\cos x$, and $f_y = e^{xy}(x\sin x + \cos x)$,

(b) $f_y = xe^{xy}\sin x$, and $f_x = e^{xy}(\sin x - y\cos x)$,

(c) $f_y = ye^{xy}\cos x$, and $f_x = e^{xy}(y\sin x + \cos x)$,

(d) $f_x = ye^{xy}\sin x$, and $f_y = e^{xy}(x\cos x - \sin x)$,

(e) none of the above. [2]

Section B

Enter your full answers in the answer booklet. This section has three questions, each worth 20 marks.

1. (a) Obtain all complex solutions of the equation $\sin(iz) = -2$. Hence, or otherwise, obtain all complex solutions of $\sinh^2(z) = -4$. [Hint: see also Section A, Q1.] [7]

   (b) Given $g(x) = \dfrac{e^{-x}\ln x}{x}$, obtain all stationary points and their nature, and show that the interval $[1,2]$ contains a stationary point. Obtain all asymptotes, intercepts, points of inflection, and hence sketch $g(x)$, explaining your reasoning. [Hint: See also Section A, Q5&6.] [7]

   (c) Given $g(x) = \dfrac{e^{-x}\ln x}{x}$ as in Q1b, let $f(x) = g(x)[H(x) - H(x-2)]$, where $H(x)$ is the Heaviside function. Sketch $f(x)$, and hence sketch $f_e(x)$ and $f_o(x)$, the even and odd parts of $f(x)$, taking care to label the axes. [6]

2. (a) Obtain the integrals: $\displaystyle\int \sinh^{-1} x \, dx$ and $\displaystyle\int \frac{1}{(1+x^2)^2}\, dx$. [7]

   (b) Find a recurrence relation for the integral

   $$I_n = \int_0^{\pi/2} \sin^n(2x)\, dx ,$$

   and use this to obtain I_4 and I_5. [6]

   (c) Let a linear, homogenous second-order ODE with constant coefficients

   $$\frac{d^2x}{dt^2} + A\frac{dx}{dt} + Bx = 0 \qquad (*)$$

   have A, B real. Let $y_1(t)$ and $y_2(t)$ be real functions of t, and $i = \sqrt{-1}$. Prove that if $z = y_1 + iy_2$ is a solution of $(*)$, then y_1 and y_2 are also solutions. [7]

3. (a) Use the Leibnitz-Maclaurin method to find the first four non-zero terms of the series solution of the second-order ODE

$$(1+t^2)\frac{d^2x}{dt^2}+2t\frac{dx}{dt}=0 \qquad (*)$$

which satisfies the initial conditions $x(0)=x'(0)=1$. Begin by differentiating the ODE n times, and obtain a recurrence relation involving $x^{(n+2)}(0)$ and $x^{(n)}(0)$. By $x^{(k)}(0)$, we mean x differentiated k times, and then evaluated at $t=0$. [8]

(b) Consider again the ODE from part (a):

$$(1+t^2)\frac{d^2x}{dt^2}+2t\frac{dx}{dt}=0 \qquad (*)$$

Now let $z=x'(t)$ and rewrite $(*)$, as a first-order linear ODE in z. Use an integrating factor to solve this, and hence find the exact solution of $(*)$. [6]

(c) Given $x(u,v)=u^2+v^2$, $y(u,v)=2uv$, and $f(x,y)$ is an arbitrary function of x and y, show that

$$u\frac{\partial f}{\partial u}-v\frac{\partial f}{\partial v} \;=\; 2(u^2-v^2)\frac{\partial f}{\partial x}$$

and find a similar expression involving $\dfrac{\partial f}{\partial y}$. [6]

MATHEMATICAL FORMULAE

1 Vector Algebra

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k} \;=\; (a_1, a_2, a_3)$$

Scalar(Dot) Product $\quad \mathbf{a}\cdot\mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$

Vector (Cross) Product

$$\mathbf{a}\times\mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2)\mathbf{i} + (a_3b_1 - a_1b_3)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}.$$

Triple vector product $\quad \mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = (\mathbf{a}\cdot\mathbf{c})\mathbf{b} - (\mathbf{a}\cdot\mathbf{b})\mathbf{c}$

Triple scalar product

$$\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) = \mathbf{b}\cdot(\mathbf{c}\times\mathbf{a}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2 - a_1b_3c_2 - a_2b_1c_3 - a_3b_2c_1$$

2 Series

$$(1+x)^n = 1+nx+\frac{n(n-1)}{2!}x^2+\frac{n(n-1)(n-2)}{3!}x^3+\ldots+\frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}x^r+\ldots \;(n\,\text{arbitrary}\,, |x|<1)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \ldots + \frac{x^r}{r!} + \ldots \qquad \text{for all } x$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots + (-1)^{r+1}\frac{x^r}{r} + \ldots \qquad \text{for } -1 < x \leq 1$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots + \frac{(-1)^r x^{2r}}{(2r)!} + \ldots \qquad \text{for all } x$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots + \frac{(-1)^r x^{2r+1}}{(2r+1)!} + \ldots \qquad \text{for all } x$$

3 Trigonometric Identities and Hyperbolic Functions

$$\sin(A+B) = \sin A\cos B + \sin B\cos A\,, \qquad \cos(A+B) = \cos A\cos B - \sin A\sin B$$

$$\cos(iz) = \cosh z\,, \quad \cosh(iz) = \cos z\,, \quad \sin(iz) = i\,\sinh(z)\,, \quad \sinh(iz) = i\,\sin z$$

$$e^{ix} = \cos(x) + i\sin(x)$$

$$\cos(x)\cos(y) = \frac{1}{2}\cos(x-y) + \frac{1}{2}\cos(x+y)$$

$$a\cos x + b\sin(x) = c\cos(x+\theta) \quad \text{where} \quad c = \sqrt{a^2+b^2} \quad \text{and} \quad \theta = \tan^{-1}\left(-\frac{b}{a}\right)$$

4 Differential Calculus

1. Leibnitz' Formula: $D^n(fg) =$

$$D^n f\, g + \binom{n}{1} D^{n-1} f\, Dg + \binom{n}{2} D^{n-2} f\, D^2 g + \cdots + \binom{n}{r} D^{n-r} f\, D^r g + \cdots + f D^n g$$

where $\left(\begin{array}{c} n \\ r \end{array} \right) = \dfrac{n(n-1)(n-2)\cdots(n-r+1)}{r!}$.

2. Taylor expansion of $f(x)$ about the point $x = a$

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x)$$

where

$$R_n(x) = \frac{(x-a)^{n+1}}{(n+1)!} f^{(n+1)}(c), \qquad a < c < x$$

Putting $x = a + h$ gives the form

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \frac{h^3}{3!} f'''(a) + \ldots$$

3. Taylor expansion of $f(x, y)$ about the point (a, b):

$$f(x,y) = f(a,b) + \frac{1}{1!}\left[(x-a)\frac{\partial}{\partial x} + (y-b)\frac{\partial}{\partial y}\right] f(a,b) + \frac{1}{2!}\left[(x-a)\frac{\partial}{\partial x} + (y-b)\frac{\partial}{\partial y}\right]^2 f(a,b) + \cdots$$

4. Chain rule for partial derivatives: $f = f(x, y)$

(i) If $x = x(t)$ and $y = y(t)$ then $f = F(t)$ and

$$\frac{dF}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

(ii) If x and y are expressed in terms of new variables u and v by $x(u, v)$ and $y(u, v)$ then $f = F(u, v)$ and

$$\frac{\partial F}{\partial u} = \frac{\partial f}{\partial x}\,\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\,\frac{\partial y}{\partial u}, \quad \text{and} \quad \frac{\partial F}{\partial v} = \frac{\partial f}{\partial x}\,\frac{\partial x}{\partial v} + \frac{\partial f}{\partial y}\,\frac{\partial y}{\partial v}$$

(iii) If $y = y(x)$ then $f = F(x)$ and

$$\frac{dF}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{dy}{dx}$$

5. Stationary points of $f(x, y)$ occur where $f_x = 0$ and $f_y = 0$. Let (a, b) be a stationary point, and consider $D = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2$. Then

(a) If $D > 0$ and $f_{xx}(a, b) < 0$, then (a, b) is a maximum;

(b) If $D > 0$ and $f_{xx}(a, b) > 0$, then (a, b) is a minimum;

(c) If $D < 0$ then (a, b) is a saddle point;

(d) if $D = 0$, more information is required.

6. Differential Equations

(i) The first order linear equation $\dfrac{dy}{dx} + p(x)y = r(x)$ has an integrating factor $I(x) = \exp\left\{\displaystyle\int p(x)\, dx\right\}$ which allows one to write the DE in the form $\dfrac{d}{dx}(I(x)y) = I(x)r(x)$.

(ii) The first-order equation $P(x, y)\dfrac{dy}{dx} + Q(x, y) = 0$ is exact if $\dfrac{\partial P}{\partial x} = \dfrac{\partial Q}{\partial y}$.

5 Integral Calculus

1. An important substitution: $\tan(\theta/2) = t$; then

$$\sin\theta = \frac{2t}{1+t^2}\,, \quad \cos\theta = \frac{1-t^2}{1+t^2}\,, \quad d\theta = \frac{2\,dt}{1+t^2}\,.$$

2.
$$\int \frac{dx}{(a^2-x^2)^{1/2}} \;=\; \sin^{-1}\left(\frac{x}{a}\right)\,, \quad |x| < a\,.$$

3.
$$\int \frac{dx}{(a^2+x^2)^{1/2}} \;=\; \sinh^{-1}\left(\frac{x}{a}\right) \;=\; \ln\left[\frac{x}{a} + \left(\frac{x^2}{a^2}+1\right)^{1/2}\right]\,.$$

4.
$$\int \frac{dx}{(x^2-a^2)^{1/2}} \;=\; \cosh^{-1}\left(\frac{x}{a}\right) \;=\; \ln\left[\frac{x}{a} + \left(\frac{x^2}{a^2}-1\right)^{1/2}\right]\,.$$

5.
$$\int \frac{dx}{a^2+x^2} \;=\; \left(\frac{1}{a}\right)\tan^{-1}\left(\frac{x}{a}\right)\,.$$

6 Fourier Series

The following formulae assume that $f(x)$ satisfies the Dirichlet conditions and is periodic with period T, i.e. $f(x+T) = f(x)$. The general Fourier Series for $f(x)$ is

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}\left[a_n\cos\left(\frac{2n\pi x}{T}\right) + b_n\sin\left(\frac{2n\pi x}{T}\right)\right]$$

where

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} f(x)\;\cos\left(\frac{2n\pi x}{T}\right)\,dx\,,\; b_n = \frac{2}{T}\int_{-T/2}^{T/2} f(x)\;\sin\left(\frac{2n\pi x}{T}\right)\,dx\,,\; n = 0,1,2,3,\ldots$$

The series converges to $f(x)$ at points of continuity and to the mean value $\frac{1}{2}(f(x_+) + f(x_-))$ at points where $f(x)$ is discontinuous.

The complex form of the Fourier series is

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i2\pi nx/T}\,, \text{ where } c_n = \frac{1}{T}\int_{-T/2}^{T/2} f(x)e^{-i2n\pi x/T}\,dx \text{ for every integer } n.$$

Parseval's Theorem: If the complex Fourier series of the T-periodic function $f(x)$ has coefficients c_n, and the real Fourier series of $f(x)$ has coefficients a_n and b_n, then

$$\frac{1}{T}\int_{-T/2}^{T/2} f^2(x)\,dx \;=\; \sum_{n=-\infty}^{\infty}|c_n|^2\,. \qquad \text{and} \qquad \frac{2}{T}\int_{-T/2}^{T/2} f^2(x)\,dx = \frac{1}{2}a_0^2 + \sum_{n=1}^{\infty} a_n^2 + b_n^2\,.$$

7 Fourier Transforms

The following formulae assume that $g(t)$ and $f(t)$ are the sum of one or more components that obey the Dirichlet conditions. The Fourier transform pair is

$$G(\omega) = \int_{-\infty}^{\infty} g(t) e^{-i\omega t}\, dt\,, \quad \Longleftrightarrow \quad g(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\omega) e^{i\omega t}\, d\omega\,,$$

where $G(\omega) = \mathcal{F}\left[g(t)\right]$ is the *Fourier transform* of $g(t)$, while $g(t) = \mathcal{F}^{-1}\left[G(\omega)\right]$ is the *inverse Fourier transform* of $G(\omega)$.

The Dirac delta function $\delta(t)$ satisfies $\displaystyle\int_{-\infty}^{\infty} \delta(t)dt = 1, \qquad \int_{-\infty}^{\infty} f(t)\delta(t-a)dt = f(a)$

For a periodic function $f(t)$ with angular frequency ω_0, and complex Fourier series as given,

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{in\omega_0 t} \qquad \Longrightarrow \qquad \mathcal{F}\left[f(t)\right] = F(\omega) = 2\pi \sum_{n=-\infty}^{\infty} F_n \delta(\omega - n\omega_0)\,.$$

Time/Frequency-shift: If $\mathcal{F}\left[f(t)\right] = F(\omega)$ then $\mathcal{F}[f(t-t_0)] = e^{-i\omega t_0} F(\omega)$ and $\mathcal{F}\left[e^{i\omega_0 t} f(t)\right] = F(\omega - \omega_0)\,.$

Symmetry/Duality property: $\mathcal{F}\left[g(t)\right] = G(\omega) \Longleftrightarrow \mathcal{F}\left[G(t)\right] = 2\pi g(-\omega)$

$\mathcal{F}\left[\cos(\omega_0 t)\right] = \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$

8 Laplace Transforms

$f(t)$	$\mathcal{L}[f(t)] = \bar{f}(s)$
1	$\dfrac{1}{s}\,, \quad \mathrm{Re}(s) > 0$
$t^n\,, \quad n = 1, 2, 3, \ldots$	$\dfrac{n!}{s^{n+1}}\,, \quad \mathrm{Re}(s) > 0$
$\sin(at)$	$\dfrac{a}{s^2 + a^2}\,, \quad \mathrm{Re}(s) > 0$
$e^{at} f(t)$	$\bar{f}(s - a)$
$\dfrac{df}{dt}$	$s\bar{f}(s) - f(0)$
$\delta(t - a)$	$e^{-as}\,, \quad a > 0$

$f(t)$	$\mathcal{L}[f(t)] = \bar{f}(s)$
e^{at}	$\dfrac{1}{s - a}\,, \quad \mathrm{Re}(s) > a$
$H(t - a)$	$\dfrac{e^{-as}}{s}\,, \quad \mathrm{Re}(s) > 0$
$\cos(at)$	$\dfrac{s}{s^2 + a^2}\,, \quad \mathrm{Re}(s) > 0$
$f(t - a)H(t - a)$	$e^{-as}\bar{f}(s)$
$\dfrac{d^2 f}{dt^2}$	$s^2\bar{f}(s) - sf(0) - \dfrac{df}{dt}(0)$
$\displaystyle\int_0^t f(u)g(t-u)\, du$	$\bar{f}(s)\bar{g}(s)$